

place_io

NAME

place_io
Places I/O driver cell instances.

SYNTAX

```
status place_io
  [-io_guide io_guide_list]
  [-rule rule_name]
  [-matching_types matching_type_list]
  [-pad_assignment_file filename]
  [-include_unassigned_pads]
  [-incremental]
  [-skip_bump_assignment]
  [-bump_assignment_only]
  [-match_terminals_to_bumps]
```

Data Types

io_guide_list	list
rule_name	string
matching_type_list	list
filename	string

ARGUMENTS

-io_guide io_guide_list
Specifies the I/O guides in which to place I/O driver cell instances. Only drivers assigned to the specified I/O guides will be placed. Other drivers will be ignored and remain in their original locations. If no driver cells are assigned to the specified I/O guides, the command will do nothing. If this option is not specified, the command will place I/O drivers in all I/O guides as well as drivers not assigned to any I/O guide.

-rule rule_name
Specifies a single routing spacing rule for pathlines. If no rule is specified, the place_io command retrieves or creates a rule that satisfies the rules set for all RDL nets. If no rule is set on any RDL nets, the spacing rule is derived from the technology file. This option is used only for flip-chip designs.

-matching_types matching_type_list
Specifies the matching types used during I/O placement. If this option is specified, only I/O driver cell instances specified in the matching types will be placed. Other drivers will be ignored and remain in their original locations. If this option is not specified, the command uses all existing matching types and places drivers not part of any matching type as well.

This option is only used for flip-chip designs since matching types describe relationship between bumps and pad pins. There exist two special types of matching types. The first type contains only bumps. These bumps are excluded from bump assignment and are not connected to any pad pins unless they have been connected to bumps in the original netlist by 2-pin nets. The second type of matching types does not contain any bump. The pads in such type of matching types will be excluded from I/O placement. They remain at their current position and are not processed during bump assignment. Their RDL pins will not be connected to any bumps unless they have been connected to bumps in the original netlist by 2-pin nets. If a matching type contains both bumps and pad pins, the place_io command assigns bumps to pad pins according to the unify number. Excessive bumps or pad pins will remain unassigned.

When this option is selected, logical connectivity of pads that are not part of any specified matching types is not changed unless such pads are connected to bumps contained in the specified matching types. Under such a situation, the pads are disconnected from the bumps.

The creation of matching types enables bump assignment for macros whose design_type attributes are set as macro, not as pad or flip_chip_driver. Normally, because macro cell instances are not considered as I/O cells, no bumps are assigned to their RDL pins. To perform bump assignment for a macro, you must create a matching type that contains the RDL macro pins and bumps to be

assigned to these pins. With such a matching type created, the `place_io` command assigns the bumps in the matching type to the pins in the matching type, with the ratio defined by the matching type `uniquify` number. User can also simply include the macro instance in the matching type. It is equivalent to include all macro RDL pins in the matching type. If user wants the bump to be connected to a specific macro pin terminal, the terminal must be included in the matching type.

When a pin has multiple terminals, only one is connected to the bump assigned. To force the tool to connect all, you must create a matching type with all terminals and the bump. In addition, the `uniquify` number of the matching type need to be set to the ratio of terminals to bumps.

-pad_assignment_file filename

Specifies allowable I/O guides for one or more I/O pads. Without such constraints, pads not assigned to I/O guides are placed within an I/O guide determined by the `place_io` command. You can limit the I/O guide candidates to a subset of all I/O guides.

The format of the constraint file is as follows. Each line contains one or more pad names followed by a list of I/O guide names. I/O guides not in the list cannot be used to place the pads in the line. If more than one pad is specified, they must be placed in a pair of curly braces. No curly braces are needed for I/O guides. Each line must end with a semicolon.

```
{pad1 pad2} guide1 guide2;  
pad3 guide1 guide3 guide0;
```

-include_unassigned_pads

Assigns any unassigned or unconstrained I/O driver cells to an I/O guide and places the cells. If this option is not specified, the command does not place the unassigned or unconstrained drivers, unless you omit both the `-io_guide` and `-matching_types` options.

-incremental

Places I/O driver cells according to their spacing and offset attributes, only for nonflip-chip designs with no I/O constraints. The `place_io` command does not consider I/O guide dimension or bump cell distribution. I/O constraints are ignored with this option.

-skip_bump_assignment

If this option is used, the tool places I/O pads only without performing bump assignment. This option and option `"-bump_assignment_only"` are mutually exclusive.

-bump_assignment_only

If this option is used, the tool does only bump assignment. No pad placement is done and I/O driver cells are not moved. The tool assigns bumps to I/O driver cells based on current locations of I/O driver cells. All the signal I/O constraints and power I/O constraints will be ignored in this mode. This option does not work with `-pad_assignment_file` option. This option and option `"-skip_bump_assignment"` are mutually exclusive.

-match_terminals_to_bumps

Matches each terminal of a pin to a bump cell. This option only works for pin terminals that fit the following requirements:

- a. The label of the pin terminal must be "bump".
- b. There is no one-to-one logical connection between the pin and a bump.

If a pin has multiple terminals fitting these requirements, the tool assigns multiple bumps to the pin. For a pin that does not meet these requirements, the tool assigns bumps to the pin in the default way.

By default, the tool picks only one terminal of a pin that has the "bump" label and assigns one bump to the pin during I/O placement.

DESCRIPTION

This command places I/O driver cell instances in I/O guides. By default, I/O driver cell instances that are not assigned to an I/O guide are automatically assigned to available I/O guides and placed. All bumps are considered.

If any pad has a fixed status, it is not moved. In addition, you can also set signal I/O constraints to specify pad orders, pad locations, and space between pads. It is your responsibility not to introduce any constraint conflicts. For example, if the space between two fixed pads is described in a signal I/O constraint, the value must be equal to the actual space between the two fixed pads.

The orientation of a pad is determined by the orientation of its reference cell and the I/O guide which the pad is placed in. Specifically, if the pad is placed in an I/O guide with side set to bottom, its orientation is the same as that of the reference cell. If the pad is placed in I/O guides with different side values, the pad will be rotated in the same way as that of I/O guide with respect to a bottom guide. You can change the reference cell orientation by setting the `reference_orientation` attribute of the reference cell. Consequently, pad orientations will be changed accordingly. For example, if users set the `reference_orientation` to MY, a pad placed in a bottom guide will be y-axis flipped. The following command sets the `reference_orientation` to R270 for all pads whose `ref_block` attribute is the same as that of `pad1`. As a result, those pads will be rotated by 270 degrees clockwise.

This command ignores non-I/O cells. This may cause an overlap between non-I/O cells and I/O pads. In case, such an overlap needs to be avoided the user can create matching types including all and only such non I/O cells. The command will then treat all such non I/O cells as regular I/O pads and fix their position.

This command will delete all the unfixed fillers in the design.

```
prompt> set_attribute -name reference_orientation \  
[ get_attribute -name ref_block [get_cells pad1]] -value R270
```

You can flip a pad along the direction perpendicular to the I/O guide of the pad. Specifically, you can set the `is_flipped` attribute as follows. This attribute can only be set on I/O cells placed during I/O placement.

```
prompt> set_attribute -name is_flipped -value true PAD1
```

EXAMPLES

The following example places I/O drivers.

```
prompt> place_io
```

SEE ALSO

[create_io_guide\(2\)](#)
[create_matching_type\(2\)](#)
[set_power_io_constraints\(2\)](#)
[set_signal_io_constraints\(2\)](#)
[write_io_constraints\(2\)](#)

Version V-2023.12-SP5-6

Copyright (c) 2025 Synopsys, Inc. All rights reserved.